

1.) Identify your problem statement:

Based on numeric output and database input

Machine Learning -> Supervised Learning -> Regression

2) the dataset (Total number of rows, columns)

Total number of rows: 1338

Columns: 6

Inputs: Age, Sex, Bmi, children, smoker

Outputs: charges

3) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

Ordinal method – used to convert for categorical data like sex, smoker

4)

Simple Linear Regression – Only one input. Here multiple input. So it is not under simple linear regression.

Multiple Linear Regression – R2 score: 0.7894790349867009

SVM:

S.NO	kernel	Hyper Parameter c	R2_score
1	linear	1	-0.01010266532
2	linear	100	0.6288792857
3	linear	1000	0.7649311739
4	poly	1	-0.07569965571
5	poly	100	0.6179569624
6	poly	1000	0.8566487676
7	rbf	1	-0.08338238594
8	rbf	100	0.3200317832

9	rbf	1000	0.8102064852
10	sigmoid	1	-0.07542924281
11	sigmoid	100	0.5276103547
12	sigmoid	1000	0.2874706949

SVM high R2_Score = 0.8566487676

Decision Tree:

S.NO	Criterion	Splitter	Max_features	R2_score
1	squared_error	Best	None	0.7140816304
2	squared_error	Best	sqrt	0.7500676078
3	squared_error	Best	log2	0.7549568247
4	squared_error	random	None	0.6943339659
5	squared_error	random	sqrt	0.772413256
6	squared_error	random	log2	0.6306134389
7	friedman_mse	Best	None	0.7014459776
8	friedman_mse	Best	sqrt	0.7710669619
9	friedman_mse	Best	log2	0.623182294
10	friedman_mse	random	None	0.7179747546
11	friedman_mse	random	sqrt	0.6150078353
12	friedman_mse	random	log2	0.710627425
13	absolute_error	Best	None	0.6872093545
14	absolute_error	Best	sqrt	0.7309290032
15	absolute_error	Best	log2	0.6897381425
16	absolute_error	random	None	0.7105340928

17	absolute_error	random	sqrt	0.7295681991
18	absolute_error	random	log2	0.7035024153
19	poisson	Best	None	0.7356102699
20	poisson	Best	sqrt	0.7128382188
21	poisson	Best	log2	0.7135545487
22	poisson	random	None	0.7376020114
23	poisson	random	sqrt	0.6164654102
24	poisson	random	log2	0.7236619797

Decision Tree highest r2_score = 0.772413256

Random Forest:

S.NO	Criterion	Max_features	R2_score
1	squared_error	None	0.8569653591
2	squared_error	1	0.852903578
3	squared_error	sqrt	0.8735638106
4	squared_error	log2	0.8708513121
5	friedman_mse	None	0.8523674626
6	friedman_mse	1	0.8565767929
7	friedman_mse	sqrt	0.8717267662
8	friedman_mse	log2	0.8670323329
9	absolute_error	None	0.8545942671
10	absolute_error	1	0.8586179217
11	absolute_error	sqrt	0.8724309696

12	absolute_error	log2	0.8732578442
13	poisson	None	0.8547727894
14	poisson	1	0.8569655167
15	poisson	sqrt	0.8687476357
16	poisson	log2	0.8721123878

Random Forest highest r2_score = 0.8735638106

5) Mention your final model, justify why u have chosen the same.

Multiple Linear Regression – R2 score: 0.7894790349867009

SVM high R2_Score = 0.8566487676

Decision Tree highest r2_score = 0.772413256

Random Forest highest r2_score = 0.8735638106

Based on above R2_score: Random Forest is highest r2_score

Because, r2_score is highest compared to other model.